Oracle Utilities Customer Care and Billing v25.10HOT

Product Fix Design

Bug 39234264

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Oracle Utilities Customer Care and Billing Product Fix Design

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Reported Issue

Description

The ‘Freeze and Complete Pending Bills’ batch (C1-BLCMP) repeatedly hangs while processing a large account that was rebilled and cancelled multiple times while updating MDM’s usage transaction on bill completion.

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Steps To Recreate

Solution Design

To address the C1-BLCMP performance issue, the BillCompletionUpdateMDMUsageInfoFromCobolWrapper flow should be changed so that it does not invoke the C1-UpdMDMUsI service script during bill completion for C2M. This script can continue to run in its separate instance where applicable.

Instead, introduce a new SQL-based approach, like the existing logic, to identify the relevant Usage Requests and derive the corresponding Usage Transactions by joining through USG_EXT_ID, which stores the Usage Request ID and is indexed on the UT side.

To reduce unnecessary updates to MDM Usage Transactions:

- For Bill Segment Cancellation (BX) financial transactions, exclude records where the Usage Transaction flags ‘Used on Completed Bill’ and ‘Linked to Frozen BS’ are already both 'N'. These are cases typically already handled when the FT freeze algorithm C1UPDMDMUFTF is plugged in.
- For Bill Segment (BS) financial transactions, exclude records where both flags are already 'Y'. This avoids redundant updates in scenarios where a completed bill is reopened and then completed again.

After identifying the applicable Usage Transactions, build the update SQL so that:

- for BX, both flags are set to 'N'
- for BS, both flags are set to 'Y'

These updates should then be performed as a bulk update. In addition, if a Usage Transaction is GDE-enabled, its UT IDs should be mass inserted as part of the processing.

The SQL should also be corrected to include CIS Division in the join between SA and SA Type, since that condition is currently missing and could lead to incorrect record matching.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	N
Is there an impact on any batch process?	**Y**
Is there an impact on CM – Custom Modification processing?	Y
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

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Other Changes

Java changes

BillCompletionUpdateMDMUsageInfoFromCobolWrapper.java

Business Logic

- If C2M
    - Retrieve UT Details.
    - For each Usage Transaction retrieved
        - If financial transaction type is Bill Segment (BS)
            - Set Used on Bill = D1YS
            - Set Lined to Frozen BS = D1YS
        - If financial transaction type is Bill Segment (BX)
            - Set Used on Bill = D1NO
            - Set Lined to Frozen BS = D1NO
    - If Generalized Data Extract is enabled for Usage Transaction
        - Register the change on the Usage Transaction Maintenance Object

          Note: This is similar the registerChange on the GDEHelper class except a list of String Ids are passed along with the Maintenance Object instead of list of Entity Ids.
- Else
    - *Existing logic……*

Retrieve UT Details

- This is like the existing method to retrieve FT Usage details except the SQL is modified to retrieve MDM's Usage Transaction with the external Id.
- To reduce unnecessary updates to MDM Usage Transactions:
    - For Bill Segment Cancellation (BX) financial transactions, exclude Usage Transactions with:
        - Used on Completed Bill = 'N' and Linked to Frozen BS = 'N'
    - For Bill Segment (BS) financial transactions, exclude Usage Transactions with:

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- Used on Completed Bill = ‘Y’ and Linked to Frozen BS = ‘Y’

SQL

```
SELECT
   FT.FT_ID,
   UT.D1_USAGE_ID
FROM
   CI_FT FT,
   CI_SA SA,
   CI_SA_TYPE SATYPE,
   C1_USAGE USG,
   D1_USAGE UT
WHERE
   FT.BILL_ID = :billId
   AND FT.FT_TYPE_FLG IN (‘BS’, ‘BX’)
   AND FT.SA_ID = SA.SA_ID
   AND SATYPE.SA_TYPE_CD = SA.SA_TYPE_CD
   AND SATYPE.CIS_DIVISION = SA.CIS_DIVISION
   AND SATYPE.SPECIAL_ROLE_FLG = ‘BD’
   AND USG.BSEG_ID = FT.SIBLING_ID
   AND USG.CRE_DTTM = (SELECT MAX(CRE_DTTM) FROM C1_USAGE WHERE BSEG_ID
= FT.SIBLING_ID)
   AND UT.USG_EXT_ID = USG.USAGE_ID
   AND FT.CRE_DTTM = (SELECT MAX(x.CRE_DTTM) FROM CI_FT x WHERE
x.SIBLING_ID = FT.SIBLING_ID AND x.BILL_ID = FT.BILL_ID)
   AND NOT ( (FT.FT_TYPE_FLG = ‘BX’ AND UT.USED_ON_BILL_FLG = 'D1NO'
AND UT.LINKED_TO_FRZN_BSEG_FLG = 'D1NO')
             OR
            (FT.FT_TYPE_FLG = ‘BS’ AND UT.USED_ON_BILL_FLG = 'D1YS'
AND UT.LINKED_TO_FRZN_BSEG_FLG = 'D1YS')
           )

ORDER BY FT.CRE_DTTM
```

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Explain Plan

OPERATION	OBJECT_NAME	OPTIONS	CARDINALITY	COST
SELECT STATEMENT			1	441
SORT		ORDER BY	1	441
FILTER				
NESTED LOOPS			1	418
NESTED LOOPS			4203	418
NESTED LOOPS			9	281
NESTED LOOPS			1	269
NESTED LOOPS			1	269
HASH JOIN			1	269
TABLE ACCESS	CFMADM.CI_FT	BY INDEX ROWID BATCHED	33	136
INDEX	CFMADM.XT112S2	RANGE SCAN	46	1
VIEW	SYS.VW_SQ_1		701	133
HASH		GROUP BY	701	133
TABLE A	CFMADM.CI_FT	BY INDEX ROWID BATCHED	701	132
INDI	CFMADM.XT112S2	RANGE SCAN	280	1
TABLE ACCESS	CFMADM.CI_SA	BY INDEX ROWID	1	1
INDEX	CFMADM.XM199P0	UNIQUE SCAN	1	0
TABLE ACCESS	CFMADM.CI_SA_TYPE	BY INDEX ROWID	1	1
INDEX	CFMADM.XC209P0	UNIQUE SCAN	1	0
TABLE ACCESS	CFMADM.C1_USAGE	BY INDEX ROWID BATCHED	38	44
INDEX	CFMADM.CT368S2	RANGE SCAN	51	1
INDEX	CFMADM.D1T419S1	RANGE SCAN	467	8
TABLE ACCESS	CFMADM.D1_USAGE	BY INDEX ROWID	1	426
SORT		AGGREGATE	1	
TABLE ACCESS	CFMADM.C1_USAGE	BY INDEX ROWID BATCHED	61	22
INDEX	CFMADM.CT368S2	RANGE SCAN	25	1

Data Clean-Up / Fixes

- N/A

Recommended Test Cases

Test Case 1: Online Bill Completion.

Generalized Data Export enabled for Usage Transaction

- Navigate to **Generalized Export Dashboard**, Enable the Ongoing Export for **Usage Transaction**.
- Create an account setup and generate bill for the account.
- Complete the bill online.

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- Verify if the Usage transaction corresponding to the financial transaction of the Bill are updated with
    - **USED_ON_BILL_FLG** – ***D1YS***
    - **LINKED_TO_FRZN_BSEG_FLG** – ***D1YS***

o Rebill the bill segment after completion and move the bill to pending. This would introduce a new bill segment and corresponding usage transaction along with the cancelled bill segment and the corresponding usage transactions.

o Repeat this for a couple of times.

o Complete the bill online again and verify if the usage transactions that satisfy the below conditions are picked for update.

- Latest FT on for the respective Bill segment is of type ***BS***
    - **LINKED_TO_FRZN_BSEG_FLG** is ***D1NO*** and/or **USED_ON_BILL_FLG** is ***D1NO***
- Latest FT on the Bill segment is of type ***BX***
    - **LINKED_TO_FRZN_BSEG_FLG** is ***D1YS*** and/or **USED_ON_BILL_FLG** is ***D1YS***

o Verify that all the updated UT's are being inserted into the following ***F1_MO_UPD*** table.

Generalized Data Export disabled for Usage Transaction

- Navigate to **Generalized Export Dashboard**, Disable the Ongoing Export for **Usage Transaction**.
- Create an account setup and generate bill for the account.
- Complete the bill online.
- Verify if the Usage transaction corresponding to the financial transaction of the Bill are updated with
    - **USED_ON_BILL_FLG** – ***D1YS***
    - **LINKED_TO_FRZN_BSEG_FLG** – ***D1YS***

o Rebill the bill segment after completion and move the bill to pending. This would introduce a new bill segment and corresponding usage transaction along with the cancelled bill segment and the corresponding usage transactions.

o Repeat this for a couple of times.

o Complete the bill online again and verify if the usage transactions that satisfy the below conditions are picked for update.

- Latest FT on for the respective Bill segment is of type ***BS***
    - **LINKED_TO_FRZN_BSEG_FLG** is ***D1NO*** and/or **USED_ON_BILL_FLG** is ***D1NO***

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    - Latest FT on the Bill segment is of type ***BX***
        - **LINKED_TO_FRZN_BSEG_FLG** is ***D1YS*** and/or **USED_ON_BILL_FLG** is ***D1YS***

- Verify that the UT's are not being inserted into the following ***F1_MO_UPD*** table.

Test Case 2: Run Batch for Freeze and Complete Pending Bills C1-BLCMP.

Generalized Data Export enabled for Usage Transaction

- Navigate to **Generalized Export Dashboard**, Enable the Ongoing Export for **Usage Transaction**.
- Create an account setup and generate bill for the account.
- Complete the bill using batch C1-BLCMP.
- Verify if the Usage transaction corresponding to the financial transaction of the Bill are updated with
    - **USED_ON_BILL_FLG** – ***D1YS***
    - **LINKED_TO_FRZN_BSEG_FLG** – ***D1YS***

- Rebill the bill segment after completion and move the bill to pending. This would introduce a new bill segment and corresponding usage transaction along with the cancelled bill segment and the corresponding usage transactions.
- Repeat this for a couple of times.
- Complete the bill again using ***C1-BLCMP*** batch and verify if the usage transactions that satisfy the below conditions are picked for update.
    - Latest FT on for the respective Bill segment is of type ***BS***
        - **LINKED_TO_FRZN_BSEG_FLG** is ***D1NO*** and/or **USED_ON_BILL_FLG** is ***D1NO***
    - Latest FT on the Bill segment is of type ***BX***
        - **LINKED_TO_FRZN_BSEG_FLG** is ***D1YS*** and/or **USED_ON_BILL_FLG** is ***D1YS***

- Verify that all the updated UT's are being inserted into the following ***F1_MO_UPD*** table.

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Generalized Data Export disabled for Usage Transaction

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- Create an account setup and generate bill for the account.
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o Verify that the UT's are not being inserted into the following ***F1_MO_UPD*** table.

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